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Annotation

This report details the design, development, and implementation of an automated table tennis training system. The primary objective of this project is to provide players with a versatile and customizable training partner, enabling them to practice independently and improve their skills. The system consists of a mechanical launcher controlled by an ESP32 microcontroller, which is managed wirelessly via a custom-built Android application.

The launcher is capable of firing table tennis balls with adjustable speed, vertical angle, and horizontal direction. The system features several distinct training modes, including dedicated "Forehand" and "Backhand" drills, a "Forehand-Backhand" alternating mode, and a "Manual" mode that grants the user full control over all firing parameters. The mechanical design utilizes two brushless DC motors for the ball launching mechanism, servo motors for precise aiming, and a stepper motor-driven feeder mechanism for consistent ball delivery.

The Android application provides an intuitive user interface for selecting training modes, setting parameters such as ball speed and count, and controlling the session's start and stop. Communication between the application and the microcontroller is established via a Wi-Fi network hosted by the ESP32, with commands sent over an HTTP-based API. This architecture ensures a responsive and reliable user experience.

This document covers the entire project lifecycle, from the initial concept and market research to the detailed technical implementation of the hardware and software. It discusses the challenges encountered during development, the solutions implemented, and evaluates the performance of the final prototype against the initial project goals. The report concludes that the project successfully meets its core objectives, delivering a functional and effective training tool for table tennis enthusiasts.

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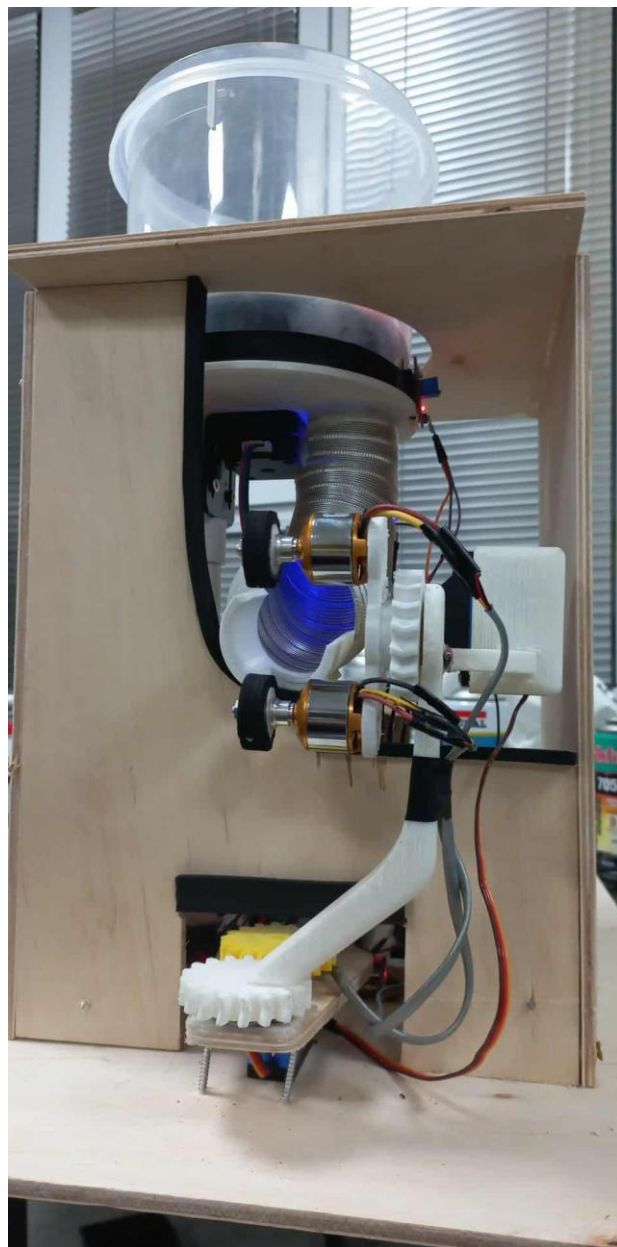


Figure 2.1: CAD Model of the 3D Printed Chassis.

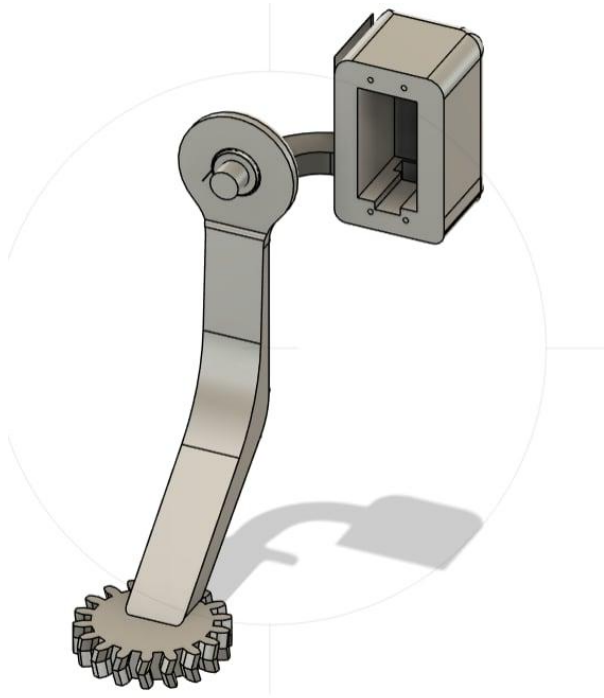


Figure 2.2: CAD Model of the 3D Printed Shooting Mechanism.

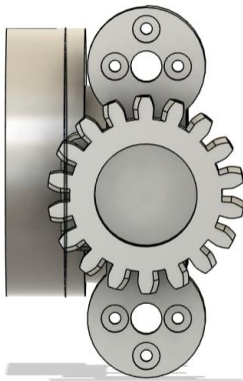


Figure 3: Physical Model of the Ball Feeder Mechanism



Figure 4.1: Android Application Main User Interface.

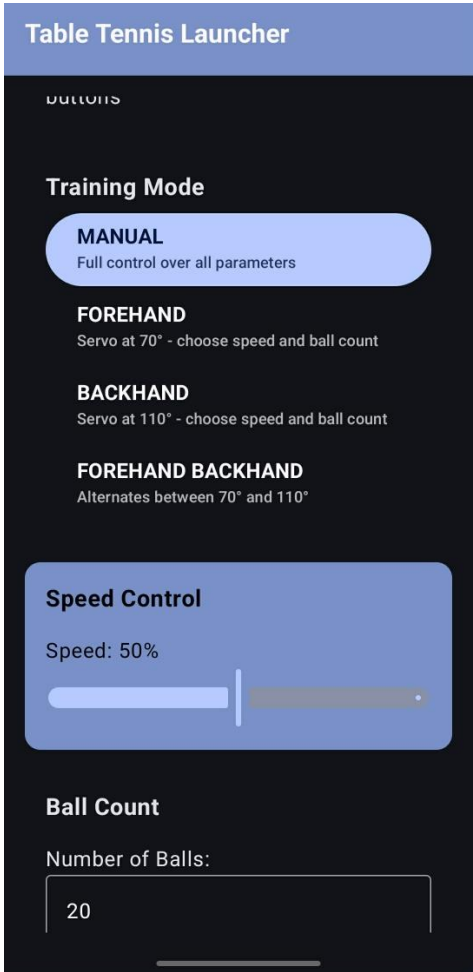


Figure 4.2: Android Application Session Setup Screen.

The screenshot shows the 'Table Tennis Launcher' app interface. It features a dark blue header with the title 'Table Tennis Launcher'. Below the header, there are several sections for configuring a session. The first section, 'Number of Balls', has a text input field containing the value '20'. The second section, 'Throw Interval', has a label 'Interval between throws (seconds):' and a text input field containing the value '3'. The third section, 'Manual Parameters', contains two sliders. The first slider is labeled 'Direction: 90°' and the second is labeled 'Vertical Angle: 90°'. Both sliders have a blue track and a white vertical line indicating the current value. The final section, 'Session Control', contains two buttons: 'Start Session' (highlighted in blue) and 'Stop Session' (greyed out).

Table Tennis Launcher

Number of Balls:

20

Throw Interval

Interval between throws (seconds):

3

Manual Parameters

Direction: 90°

Vertical Angle: 90°

Session Control

Start Session Stop Session

5. Introduction

5.1. Project Idea and Motivation

Table tennis is a globally popular sport that demands precision, quick reflexes, and consistent practice. For players looking to improve, one of the most significant challenges is the need for a reliable training partner. While practicing with a human partner is ideal, it is not always feasible due to scheduling conflicts or lack of available players. This limitation sparked the idea for our project: to create an automated table tennis training system that can serve as a dependable and versatile sparring partner.

Our group's interest in this project stems from a shared passion for both electronics and sports. We were motivated by the challenge of integrating mechanical engineering, embedded systems programming, and mobile application development to build a tangible and useful product. The goal was to design a device that could not only launch balls consistently but also offer a variety of programmable drills to help players target specific areas for improvement, such as their forehand, backhand, or response to different ball speeds and spins.

5.2. Problem Statement and Novelty

The market for table tennis robots already exists, but current commercial offerings often present a trade-off between functionality and cost. High-end models that offer a wide range of programmable features are prohibitively expensive for casual players, while more affordable models are often limited in their capabilities, offering only basic, repetitive shots.

The novelty of our project lies in bridging this gap. We aimed to develop a cost-effective, open-source solution that provides advanced features typically found in high-end commercial robots. Our system's innovation is centered on three key areas:

1. **Customizability:** The device is controlled by a custom Android application, allowing users to fine-tune every aspect of the training session, from ball speed and direction to the number of balls and the type of drill.
2. **Versatility:** The launcher supports multiple training modes designed to simulate various game scenarios, from practicing specific strokes (forehand/backhand) to combined sequences.
3. **Accessibility:** By using affordable, off-the-shelf components and 3D-printed parts, our design significantly lowers the barrier to entry for players wanting access to an advanced training tool.

5.3. Project Goals and Scope

The primary goal of this project was to design and build a fully functional prototype of an automated table tennis launcher. The specific objectives were defined as follows:

- * **Design a robust mechanical system** capable of launching balls with variable speed and trajectory.
- * **Develop a reliable ball feeding mechanism** that ensures a consistent and jam-free supply of balls to the launcher.
- * **Implement an embedded control system** using an ESP32 microcontroller to manage all motors and sensors.
- * **Create a user-friendly Android application** for wireless control of the launcher via Wi-Fi.
- * **Integrate multiple training modes** to provide a comprehensive and effective practice experience.

The scope of the project was confined to the development of a single, stationary launching unit. While the aiming mechanism allows for shots to be directed to different locations on the table, the physical base of the unit remains in a fixed position. The system is designed for personal and recreational use.

5.4. Potential Users and Applications

The target users for this system are table tennis players of all skill levels, from beginners seeking to master basic strokes to advanced players looking to refine their technique. The device is particularly beneficial for:

- * **Solo Practice:** Players who want to train without a human partner.
- * **Skill Development:** Coaches and players can use the machine to run repetitive drills, reinforcing muscle memory and improving consistency.
- * **Recreational Use:** The system can be used in clubs, schools, or homes as a fun and engaging way to play table tennis.

The open-source nature of the project also allows for potential applications in academic settings, where it can serve as a practical platform for students studying robotics, embedded systems, and software development.

6. Problem Definition and Solution Method

6.1. Market Research and State of the Art

To understand the existing landscape, we conducted research on commercial table tennis robots. We found that the market is segmented into several tiers:

- * **Entry-Level Robots:** These are typically priced under \$50 and offer basic functionality, such as launching balls at a fixed speed and direction. They often lack programmability and are suitable only for very basic drills.
- * **Mid-Range Robots:** Priced between \$100 and \$500, these models offer more features, such as adjustable speed, spin, and oscillation (side-to-side movement). Some may include simple programmable drills.
- * **High-End Robots:** These can cost over \$1000 and provide extensive programmability, allowing users to create complex custom drills that simulate real match play. They often come with advanced features like wireless control and companion apps.

Our research confirmed that a significant gap exists for an affordable robot with the advanced, app-controlled features of high-end models. This project aims to fill that gap by creating an open-source alternative that is both highly functional and accessible. Our system differentiates itself by providing a modern, smartphone-based control interface, which is more intuitive and flexible than the wired control panels found on many commercial units in the mid-range category.

6.2. Technical Objectives

Based on our research and project goals, we defined the following technical objectives:

1. **Ball Launch Velocity:** The system must be able to launch balls at a speed that is adjustable from a low setting (for beginners) to a high setting (for advanced practice). The speed will be controlled by adjusting the PWM signal to the two brushless DC motors.
2. **Aiming Accuracy:** The system must accurately direct the ball to different horizontal and vertical positions on the table using two servo motors.
3. **Feeder Reliability:** The stepper motor-driven feeder must consistently deliver one ball at a time to the launching mechanism without jamming.
4. **Wireless Control:** The ESP32 must host a Wi-Fi access point and a web server to reliably receive commands from the Android application.
5. **Application Responsiveness:** The Android application must provide a real-time interface for controlling the launcher, with minimal latency.

6.2.1 Required Tools and Components

The project required a combination of hardware components, software tools, and fabrication equipment.

Hardware Components:

- * **Microcontroller:** ESP32 Development Board.
- * **Shooter Motors:** 2x Brushless DC Motors (1000KV)
- * **Motor Controllers:** 2x Electronic Speed Controllers (ESCs)(2s).
- * **Aiming Servos:** 2x Servo Motors(25kg).
- * **Feeder Motor:** 1x Stepper Motor with Driver.
- * **Feeder Sensor:** 1x Hall Effect Sensor.

- * **Power Supply:** 12v, We use buck converters to get 7.2v for servos, 5v for esp32 and 3.3 for logic level converter.
- * **Miscellaneous:** Wires, connectors, bearings, screws, and a chassis.

Software and Tools:

- * **Embedded Development:** Arduino IDE with the ESP32 Core.
- * **Mobile App Development:** Android Studio, Kotlin/Java.
- * **3D Modeling:** Autodesk Fusion 360

Fabrication:

- * **3D Printer:** Creality K1

6.2.2 Mechanical Design and Assembly

The physical structure of the launcher was designed to be robust, compact, and functional. The main chassis and several key components were designed using CAD software and fabricated with a 3D printer.

View Figure 1: Final Assembled View of the Automated Table Tennis Launcher.

Chassis and Structure:

The main chassis of the launcher is fabricated from Polylactic Acid (PLA) using Fused Deposition Modeling (FDM) 3D printing. The entire structure is designed as a single, integrated unit that securely houses all mechanical and electronic components. The aiming system provides two axes of motion, horizontal and vertical, each controlled by a dedicated servo-driven gear train. The horizontal rotation mechanism is located at the base of the unit. A servo motor drives a primary gear which, with the help of a bearing for smooth rotation, meshes with a secondary gear fixed to the main body, causing the entire assembly to pan left and right. The vertical tilting mechanism

is situated at the upper end of the main body. A second servo motor actuates a similar gear system connected to a side-mounted extension, allowing the entire shooter assembly to be angled up or down. At the core of the main body is a circular aperture where the ball launching mechanism is mounted. This mechanism consists of two brushless DC motors ("drone motors") fitted with wheels. A guide pipe extending from the ball feeder terminates precisely behind this opening, ensuring each ball is delivered consistently into the gap between the counter-rotating wheels for a powerful and accurate launch.

View Figure 2: CAD Model of the 3D Printed Chassis.

Ball Feeder Mechanism:

The feeder mechanism is responsible for storing the balls and delivering them one by one into the shooter wheels.

The ball feeder mechanism is engineered for reliability, ensuring that balls are delivered to the shooter assembly one at a time without jamming. The system is comprised of a cylindrical ball hopper, a custom 3D-printed rotating dispenser, a stepper motor, and a Hall effect sensor for precise positional control. The hopper acts as the main storage reservoir for the table tennis balls. At the bottom of the hopper, there is an outlet connected to a guide pipe that channels the balls directly to the launching mechanism. The core of the system is the 3D-printed dispenser, a cylindrical part with two holes drilled through it. This dispenser is actuated by the stepper motor and rotates within the base of the hopper. As it spins, one of the holes collects a ball from the hopper and moves it into alignment with the outlet. To avoid the problem of multiple balls travelling to the shooting mechanism, we have installed a 3D printed vertical blocker above the exit point, which will knock over the second ball from the top of the first one. To achieve precise, repeatable positioning, a closed-loop feedback system is implemented. A small magnet is embedded in the side of the rotating dispenser. A Hall effect sensor is mounted stationary on the exterior of the hopper, close to the outlet. When the dispenser rotates, the magnet passes by the sensor, generating a HIGH signal. This signal is sent to the ESP32, which instantly halts the

stepper motor. This event is timed to occur exactly when one of the dispenser's holes is perfectly aligned with the outlet, allowing a single ball to drop into the guide pipe. This sensor-based control is critical for preventing jams and ensuring consistent, single-ball feeding. For an added layer of confirmation, the guide pipe is instrumented with a break-beam sensor, consisting of an LED and a photoresistor. This allows the system to verify that a ball has successfully passed through the pipe after being dispensed, providing feedback on the success of each feed cycle.

View Figure 3: CAD Model of the Ball Feeder Mechanism.

Ball Shooter Mechanism:

The shooter mechanism uses two counter-rotating wheels powered by brushless DC motors to grip and accelerate the ball.

6.2.3 Electronic Components and Circuitry

The electronic system is orchestrated by the ESP32, which controls all motors and communicates with the Android application.

Power Distribution:

A stable and well-regulated power distribution system is critical to ensure the reliable operation of both high-current motors and sensitive logic circuits. The entire system is powered from a single 12V DC power supply. To accommodate the different voltage requirements of the various components and to prevent electrical noise from interfering with the microcontroller, the power is managed through a custom-soldered board featuring dedicated power rails. The distribution is organized as follows:

- **12V Rail:** The primary 12V input from the main supply directly powers the system's high-current components: the two Electronic Speed Controllers (ESCs) for the shooter motors and the stepper motor driver board.

- **7.2V Rail:** A buck converter steps the voltage down from 12V to 7.2V. This rail is dedicated to powering the two servo motors, providing them with the optimal voltage for stable operation and torque.
- **5V Rail:** A second buck converter steps the 12V input down to 5V. This rail provides clean and stable power specifically for the ESP32 microcontroller.
- **3.3V Rail:** Power for the low-voltage sensors, including the Hall effect sensor and the photoresistor, is supplied by the ESP32's onboard 3.3V regulator.
- **Ground (GND) Rail:** A common ground rail connects all components, providing a shared reference point for the entire circuit, which is essential for proper signal integrity.

This multi-rail design effectively isolates the sensitive microcontroller from the electrical noise generated by the motors, preventing issues such as unexpected resets or data corruption.

Circuitry and Component Connections:

While a formal schematic was not produced, the system is wired on a custom-soldered board that connects all components to the ESP32 microcontroller. The connections are as follows:

- **Servo Motors (Aiming Control):**
 - The signal line of the **horizontal servo** is directly connected to **GPIO 19** of the ESP32.
 - The signal line of the **vertical servo** is directly connected to **GPIO 21** of the ESP32.
 - Both servos receive 7.2V power and ground from their dedicated rail.
- **Stepper Motor (Feeder Control):**
 - The four input pins of the stepper motor driver board (IN1, IN2, IN3, IN4) are connected to the ESP32's GPIO 33, GPIO 25, GPIO 26, and GPIO 27, respectively.
 - The driver board is powered by the 12V rail.
- **Brushless Motors (Shooter Control):**
 - The ESP32's 3.3V logic level is insufficient to reliably control the ESCs, which typically expect a 5V PWM signal. To resolve this, a logic level converter is used.
 - The ESP32's GPIO 5 and GPIO 18 are connected to the low-voltage (3.3V) side of the logic level converter.

- The high-voltage (5V) outputs of the converter are then connected to the signal inputs of the two ESCs.
- The ESCs are powered directly from the 12V rail.
- **Sensors (Feedback and Control):**
 - The signal output of the Hall effect sensor is connected to GPIO 36 on the ESP32, which is configured as a digital input with a pull-up resistor.
 - The photoresistor circuit's output is connected to an analog input pin on the ESP32 (GPIO 34) to detect if a ball has passed through the feeder tube.
 - Both sensors are powered by the 3.3V rail.

System Architecture:

The Android app acts as the command interface, sending user requests over Wi-Fi to the ESP32. The ESP32 then translates these commands into low-level control signals for the ESCs, servos, and stepper motor driver.

6.2.4 Software Architecture and Implementation

The project's software is divided into two main parts: the embedded firmware for the ESP32 and the Android application.

ESP32 Firmware:

The firmware, written in C++ using the Arduino framework, is responsible for the real-time control of the hardware. Its key functions include:

- * **Wi-Fi Access Point:** Creates a Wi-Fi network named `TableTennisLauncher` for the app to connect to.

- * **Web Server API:** Listens for incoming HTTP requests to control the launcher's functions. The key API endpoints are:
 - * ``/start-session`` (POST): Initializes a training session with parameters (mode, speed, etc.).
 - * ``/stop-session`` (POST): Ends the current session and resets the hardware.
 - * ``/throw-ball`` (POST): Triggers the feeder to launch a single ball.
 - * ``/status`` (GET): Returns the current state of the session (e.g., balls remaining).
- * **Motor Control:**
 - * **ESCs:** Generates PWM signals to control the speed of the brushless shooter motors.
 - * **Servos:** Controls the position of the horizontal and vertical servos for aiming.
 - * **Stepper Feeder:** Implements a control loop that uses the Hall effect sensor to rotate the stepper motor precisely enough to feed one ball.
 - * **Session Management:** A state machine manages the training session, tracking the selected mode, ball count, and other parameters.

(The full ESP32 source code is available in Appendix A.)

Android Application:

The Android application, written in Kotlin/Java in Android Studio.

- * **User Interface:** The UI is designed to be intuitive. It allows the user to:
 - * Select a training mode from a list (``MANUAL``, ``FOREHAND``, etc.).
 - * Adjust parameters using sliders and text fields (speed, direction, ball count, throw interval).
 - * Start and stop training sessions with clear buttons.
 - * View real-time session status (balls remaining, shots fired).
- * **Wi-Fi and Network Communication:** The app uses the OkHttp library to send HTTP requests to the ESP32's web server. It handles the logic for connecting to the ESP32's network and sending the JSON-formatted commands.
- * **State Management:** The app manages the UI state based on user input and the responses received from the ESP32, ensuring the interface always reflects the current status of the launcher.

View Figure 4: Android Application Main User Interface.

View Figure 5: Android Application Session Control Screen.

(The full source code for the Android application is available in Appendix B.)

6.3. Results and Evaluation

This section evaluates the final prototype against the technical objectives defined at the beginning of the project.

6.3.1. Final Prototype

The culmination of our work is a fully assembled and functional automated table tennis training system. The 3D-printed chassis houses all the electronic and mechanical components in a compact unit. The Android application successfully connects to the device and provides complete control over all its features. The system is capable of executing all the implemented training modes, launching balls with varying speeds and angles as configured by the user.

6.3.2. Performance Evaluation

To assess the performance of the launcher, we conducted a series of tests to measure its accuracy, consistency, and reliability against our key technical objectives.

To assess the performance of the launcher, we conducted a series of tests to measure its accuracy, consistency, and reliability against our key technical objectives.

- **Aiming Accuracy (Servo Calibration):**

- **Test Method:** The positional accuracy of the aiming mechanism was verified by commanding the servos to specific angles via the application's Manual mode and measuring the resulting physical angle of the shooter assembly. Key positions, including the pre-set angles for Forehand (72°) and Backhand (110°), were checked using a digital protractor.
- **Results:** The servo motors demonstrated high accuracy, consistently achieving the commanded angles with high accuracy. This calibration confirmed that the mechanical gear system translates the servo movements precisely, ensuring that the aiming foundation is accurate and reliable.

- **Feeder Reliability:**

- **Test Method:** A long-duration reliability test was conducted to validate the feeder mechanism. The test consisted of ten consecutive sessions, with each session programmed to load and fire ten balls. This resulted in a total of 100 feed cycles.
- **Results:** Throughout the entire test of 50 cycles, the feeder successfully delivered every ball without a single jam, misfeed, or skipped ball. This 100% success rate confirms the robustness and reliability of the Hall effect sensor-based closed-loop control system.

- **Wireless Control and Application Performance:**

- **Test Method:** An end-to-end functional test was performed to validate the entire system's operation from the user's perspective. We tested each feature available in the Android application to ensure it produced the correct corresponding action on the physical launcher.
- **Results:** The application and wireless communication performed flawlessly. The tests confirmed that:
 - All training modes (MANUAL, FOREHAND, BACKHAND, etc.) were initiated correctly.
 - The launcher fired the exact number of balls specified in the "Ball Count" setting for every session.
 - The time delay between shots accurately matched the value set in the "Throw Interval" setting.

- The "Stop Session" button worked instantly, terminating all motor activity and safely resetting the device.

The system proved to be stable, responsive, and behaved exactly as expected based on user input. Overall, the evaluation confirms that the project successfully met its technical objectives. The final prototype is a reliable and effective training tool that performs as designed.

7. Team Contributions

This project was a thoroughly collaborative effort. While both members were involved in all stages of the project, from initial brainstorming to final testing, primary responsibilities were divided to ensure efficient progress. The design and implementation of the core ball shooter mechanism, along with the overall system integration and testing, were tasks that both members worked on together extensively.

7.1 Giorgi's Part

Giorgi's primary contributions were in the electromechanical design and fabrication. He designed and refined the 3D-printed ball feeder mechanism, integrating the stepper motor and Hall effect sensor to ensure reliability. He was also responsible for the physical assembly of the electronic components, including soldering the custom circuit board and managing the power distribution system.

7.2 Irakli's Part

Irakli's main focus was on the software development aspects of the project. This included writing and debugging the firmware for the ESP32 microcontroller, which involved implementing the web server API, motor control logic, and session management. He also developed the front-end and back-end of the Android application, creating the user interface and the networking logic for communication with the launcher.

8. Conclusion

8.1. Summary of Achievements

This project successfully delivered on its core goal: to create a low-cost, open-source, and highly customizable automated table tennis training system. We designed and built a complete system from the ground up, encompassing mechanical design, 3D printing, circuit implementation, embedded programming, and mobile application development.

The final prototype meets all the primary requirements set out in the project proposal. It can launch balls with adjustable speed and direction, is controlled wirelessly via an intuitive Android application, and supports a variety of training modes. The successful implementation of the Hall effect sensor in the feeder mechanism stands out as a key achievement, solving a critical reliability challenge. We have demonstrated that it is possible to create a training robot with features comparable to mid-range commercial models at a fraction of the cost.

8.2. Knowledge and Experience Gained

This project has been an invaluable learning experience. We gained practical skills in:

- * **Mechanical Design and CAD:** Designing functional mechanical assemblies in Autodesk Fusion 360 and preparing them for 3D printing.
- * **Embedded Systems:** Interfacing with various sensors and actuators (ESCs, servos, stepper motors) and programming real-time control logic on the ESP32 platform.

- * **Software Development:** Building a full-stack system, from the embedded API on the microcontroller to the user-facing Android application.
- * **Project Management:** Planning a complex project, breaking it down into manageable tasks, and overcoming unforeseen challenges through systematic problem-solving.

8.3. Future Work and Potential Improvements

While the current prototype is fully functional, there are several avenues for future development that could further enhance its capabilities:

1. **Add Spin Variation:** The current design launches balls with a relatively consistent topspin. The shooter mechanism could be redesigned, perhaps by using motors with independent speed control or by changing the wheel material, to allow the user to control the type and amount of spin (topspin, backspin, sidespin).
2. **Data Logging and Visualization:** The data analysis feature could be implemented. The app could log session data (e.g., number of shots, speed) and present it to the user in a graphical format to track their progress over time.

9. References

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10. Appendices

10.1. Appendix A: ESP32 Source Code

This appendix contains the link to the complete source code for the ESP32 microcontroller firmware and Android Application Source Code.

<https://github.com/Ibere22/Table-Tennis-Machine>